

18. Perceptron Based IRIS Classification

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
iris = load_iris()
```

```
X = iris.data
y = iris.target
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
model = Perceptron(
    max_iter=1000,
    eta0=0.1,
    random_state=42
)
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
print("Actual:")
print(y_test)
```

```
print("\nPredicted:")
print(y_pred)
```

```
print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))
```

```
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

OUTPUT:

Actual:

```
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]
```

Predicted:

```
[0 0 2 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0]
```

Accuracy:

0.4666666666666667

Confusion Matrix:

```
[[19  0  0]
 [12  1  0]
 [11  1  1]]
```

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